# Chapter 5. The Fourier Transform on R

*Schwartz space, inversion, Plancherel, Poisson summation*

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## 5.1 Elementary theory of the Fourier transform

**Motivation.** On the real line Fourier series become integrals: $\hat{f}(\xi) = \int f(x)e^{-2\pi ix\xi}dx$. On the Schwartz space $\mathcal{S}(\mathbb{R})$ of smooth rapidly decreasing functions the Fourier transform interchanges differentiation and multiplication by $x$, fixes the Gaussian, and is a bijection that preserves $L^2$ norms.

### 5.1.1 Integration of functions on the real line

$f$ is of **moderate decrease** if it is continuous and $|f(x)| \le \frac{A}{1+x^2}$ for some $A$. For such $f$, $\int_{-N}^{N} f$ is Cauchy as $N \to \infty$, and we set $\int_{-\infty}^{\infty} f = \lim_{N\to\infty} \int_{-N}^{N} f$. These functions form a vector space $\mathcal{M}(\mathbb{R})$.

**Proposition 1.1**

For $f, g$ of moderate decrease:
(i) Linearity: $\int(af + bg) = a\int f + b\int g$.
(ii) Translation invariance: $\int f(x-h)\,dx = \int f(x)\,dx$ for $h \in \mathbb{R}$.
(iii) Scaling: $\delta \int f(\delta x)\,dx = \int f(x)\,dx$ for $\delta > 0$.
(iv) Continuity: $\int |f(x-h) - f(x)|\,dx \to 0$ as $h \to 0$.

**Proof.** (i) Clear from finite intervals.

(ii) $\int_{-N}^{N} f(x-h)dx = \int_{-N-h}^{N-h} f$, and $\int_{-N-h}^{-N} f$, $\int_{N-h}^{N} f \to 0$ by the decay.

(iii) $\delta \int_{-N}^{N} f(\delta x)dx = \int_{-\delta N}^{\delta N} f$.

(iv) Given $\epsilon$, choose $N$ with $\int_{|x|\ge N} |f(x)|, \int_{|x|\ge N} |f(x-h)| < \epsilon$ for $|h| \le 1$; on $[-N, N]$, $f$ is uniformly continuous, so $\int_{-N}^{N} |f(x-h) - f(x)| < \epsilon$ for small $h$. ∎

### 5.1.2 Definition of the Fourier transform

For $f \in \mathcal{M}(\mathbb{R})$, $\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-2\pi ix\xi}\,dx$; $\hat{f}$ is bounded and continuous. (If $\hat{f}$ is also of moderate decrease, inversion and Plancherel hold, §5.1.7.)

### 5.1.3 The Schwartz space

$\mathcal{S}(\mathbb{R})$ consists of the $C^\infty$ functions $f$ with $\sup_x |x|^k |f^{(\ell)}(x)| < \infty$ for all $k, \ell \ge 0$ (**rapidly decreasing**). It is a vector space closed under differentiation and multiplication by polynomials. The **Gaussian** $e^{-ax^2}$ $(a > 0)$ belongs to $\mathcal{S}(\mathbb{R})$: its derivatives are $P(x)e^{-ax^2}$ with polynomials $P$. Smooth compactly supported functions also belong to it; the **bump function** $e^{-1/(1-x^2)}$ on $|x| < 1$ (0 elsewhere) is an example.

### 5.1.4 The Fourier transform on $\mathcal{S}$

**Proposition 1.2**

If $f \in \mathcal{S}(\mathbb{R})$, writing $f(x) \to \hat{f}(\xi)$:
(i) $f(x+h) \to \hat{f}(\xi)e^{2\pi ih\xi}$ for $h \in \mathbb{R}$.
(ii) $f(x)e^{-2\pi ixh} \to \hat{f}(\xi+h)$ for $h \in \mathbb{R}$.
(iii) $f(\delta x) \to \delta^{-1}\hat{f}(\delta^{-1}\xi)$ for $\delta > 0$.

(iv) $f'(x) \to 2\pi i\xi \hat{f}(\xi)$.
(v) $-2\pi i x f(x) \to \frac{d}{d\xi}\hat{f}(\xi)$.

**Proof.** (i)–(iii) by Proposition 1.1(ii), (iii).

(iv) Integrate by parts on $[-N, N]$: $\int_{-N}^{N} f'(x)e^{-2\pi i x\xi}dx = [f(x)e^{-2\pi i x\xi}]_{-N}^{N} + 2\pi i\xi \int_{-N}^{N} f(x)e^{-2\pi i x\xi}dx$, and the boundary term $\to 0$.

(v) Let $g(x) = -2\pi i x f(x) \in \mathcal{S}$. For $\epsilon > 0$,

$$\frac{\hat{f}(\xi+h) - \hat{f}(\xi)}{h} - \hat{g}(\xi) = \int f(x)e^{-2\pi i x\xi}\Big[\frac{e^{-2\pi i x h} - 1}{h} + 2\pi i x\Big]dx.$$

Choose $N$ with $\int_{|x|\geq N}|f|, \int_{|x|\geq N}|x||f| < \epsilon$. For $|x| \leq N$, $|\frac{e^{-2\pi i x h}-1}{h} + 2\pi i x| \leq C|h|$ uniformly (Taylor). Also $|\frac{e^{-2\pi i x h}-1}{h}| \leq 2\pi|x|$. So the difference is $\leq C'\epsilon + C''N|h| < C\epsilon$ for $|h|$ small. ∎

**Theorem 1.3**

If $f \in \mathcal{S}(\mathbb{R})$, then $\hat{f} \in \mathcal{S}(\mathbb{R})$.

**Proof.** $\hat{f}$ is bounded. By Proposition 1.2(iv), (v), $\xi^k(\frac{d}{d\xi})^\ell \hat{f}(\xi)$ is the Fourier transform of $\frac{1}{(2\pi i)^k}(\frac{d}{dx})^k[(-2\pi i x)^\ell f(x)] \in \mathcal{S}$, hence bounded. And $\hat{f}$ is $C^\infty$ by repeated (v). ∎

**Theorem 1.4**

If $f(x) = e^{-\pi x^2}$, then $\hat{f}(\xi) = f(\xi)$.

**Proof.** $F(\xi) = \hat{f}(\xi)$ has $F(0) = \int e^{-\pi x^2}dx = 1$. By Proposition 1.2(v) and $f'(x) = -2\pi x f(x)$, then (iv),

$$F'(\xi) = \int f(x)(-2\pi i x)e^{-2\pi i x\xi}dx = i\int f'(x)e^{-2\pi i x\xi}dx = i(2\pi i\xi)F(\xi) = -2\pi\xi F(\xi).$$

So $G(\xi) = F(\xi)e^{\pi\xi^2}$ has $G' = 0$ and $G = G(0) = 1$. ∎

($\int e^{-\pi x^2}dx = 1$: square it and use polar coordinates, $\int_0^{2\pi}\int_0^\infty e^{-\pi r^2} r\,dr\,d\theta = 1$.)

**Corollary 1.5**

If $\delta > 0$ and $K_\delta(x) = \delta^{-1/2}e^{-\pi x^2/\delta}$, then $\widehat{K_\delta}(\xi) = e^{-\pi\delta\xi^2}$.

**Proof.** Proposition 1.2(iii) with $\delta$ replaced by $\delta^{-1/2}$ and Theorem 1.4. ∎

As $\delta \to 0$, $K_\delta$ concentrates at 0 while $\widehat{K_\delta}$ flattens out: the uncertainty principle in miniature (§5.4).

**Definition (Good kernels on $\mathbb{R}$)**

$\{K_\delta\}_{\delta>0}$ is a family of **good kernels** as $\delta \to 0$ if (i) $\int K_\delta = 1$;
(ii) $\int |K_\delta| \leq M$;
(iii) for every $\eta > 0$, $\int_{|x|>\eta}|K_\delta| \to 0$.

**Theorem 1.6**

$\{K_\delta\}_{\delta>0}$, $K_\delta(x) = \delta^{-1/2}e^{-\pi x^2/\delta}$, is a family of good kernels as $\delta \to 0$.

**Proof.** $K_\delta > 0$ and $\int K_\delta = \widehat{K_\delta}(0) = 1$. Substituting $x = \delta^{1/2}y$, $\int_{|x|>\eta} K_\delta = \int_{|y|>\eta/\sqrt{\delta}} e^{-\pi y^2}dy \to 0$. ■

For $f, g \in \mathcal{S}(\mathbb{R})$, $(f * g)(x) = \int f(x-t)g(t)\,dt$.

**Corollary 1.7**

If $f \in \mathcal{S}(\mathbb{R})$, then $(f * K_\delta)(x) \to f(x)$ uniformly in $x$ as $\delta \to 0$.

**Proof.** $f$ is uniformly continuous: given $\epsilon$, take $R$ with $|f| < \epsilon/4$ for $|x| \geq R$, and use uniform continuity on $[-R-1, R+1]$. Choose $\eta$ with $|f(x-t) - f(x)| < \epsilon$ for $|t| < \eta$. Then

$$|(f * K_\delta)(x) - f(x)| \leq \int_{|t|<\eta} K_\delta(t)|f(x-t) - f(x)|dt + \int_{|t|\geq\eta} K_\delta(t)\, 2\sup|f|\, dt \leq \epsilon + C\int_{|t|\geq\eta} K_\delta < 2\epsilon$$

for $\delta$ small, uniformly in $x$. ■

### 5.1.5 The Fourier inversion

**Proposition 1.8 (Multiplication formula)**

If $f, g \in \mathcal{S}(\mathbb{R})$, then $\int f(x)\hat{g}(x)\,dx = \int \hat{f}(y)g(y)\,dy$.

**Proof.** $F(x,y) = f(x)g(y)e^{-2\pi ixy}$ has $|F| \leq \frac{C}{(1+x^2)(1+y^2)}$, so (Appendix, Theorem 3.1) the iterated integrals may be interchanged: $\int \Big(\int F\,dy\Big)dx = \int f(x)\hat{g}(x)dx$ and $\int \Big(\int F\,dx\Big)dy = \int \hat{f}(y)g(y)dy$. ■

**Theorem 1.9 (Fourier inversion)**

If $f \in \mathcal{S}(\mathbb{R})$, then $f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi)e^{2\pi ix\xi}\,d\xi$.

**Proof.**

*Step 1 ($x = 0$).* Let $G_\delta(x) = e^{-\pi\delta x^2}$, so $\widehat{G_\delta} = K_\delta$ (Corollary 1.5). By Proposition 1.8, $\int f(x)K_\delta(x)dx = \int \hat{f}(\xi)G_\delta(\xi)d\xi$. As $\delta \to 0$, the left side tends to $f(0)$ (Corollary 1.7, $K_\delta$ even), and the right side to $\int \hat{f}$ (dominated by $|\hat{f}|$, and $G_\delta \to 1$ uniformly on compacts). So $f(0) = \int \hat{f}(\xi)\,d\xi$.

*Step 2 (general $x$).* Apply Step 1 to $F(y) = f(y+x)$, whose transform is $\hat{f}(\xi)e^{2\pi ix\xi}$ (Proposition 1.2(i)). ■

Write $\mathcal{F}(f) = \hat{f}$ and $\mathcal{F}^*(g)(x) = \int g(\xi)e^{2\pi ix\xi}d\xi$; then $\mathcal{F}^*\mathcal{F} = I$ on $\mathcal{S}$, and $\mathcal{F}^*(g)(x) = \mathcal{F}(g)(-x)$.

**Corollary 1.10**

The Fourier transform is a bijection of $\mathcal{S}(\mathbb{R})$ onto itself.

**Proof.** $\mathcal{F}$ maps $\mathcal{S}$ to $\mathcal{S}$ (Theorem 1.3), $\mathcal{F}^* \circ \mathcal{F} = I$ (Theorem 1.9), and similarly $\mathcal{F} \circ \mathcal{F}^* = I$ (since $\mathcal{F}^*(g) = \mathcal{F}(g)(-\cdot)$). ■

### 5.1.6 The Plancherel formula

**Proposition 1.11**

If $f, g \in \mathcal{S}(\mathbb{R})$:
(i) $f * g \in \mathcal{S}(\mathbb{R})$;
(ii) $f * g = g * f$;
(iii) $\widehat{f * g}(\xi) = \hat{f}(\xi)\hat{g}(\xi)$.

**Proof.**

(ii) Substitute $t \mapsto x - t$.

(i) For $\ell \geq 0$: $\sup |x|^\ell |g(x-y)| \leq A_\ell (1+|y|)^\ell$ since $|x|^\ell \leq 2^\ell(|x-y|^\ell + |y|^\ell)$. So $\sup_x |x^\ell (f*g)(x)| \leq A_\ell \int |f(y)|(1+|y|)^\ell dy < \infty$. Derivatives: $\frac{d}{dx}(f*g) = f * g'$ (differentiate under the integral), and repeat.

(iii) $F(x,y) = f(y)g(x-y)e^{-2\pi i x\xi}$ is integrable on $\mathbb{R}^2$ (rapid decay), so by Fubini (Theorem 3.1), $\int (f*g)(x)e^{-2\pi i x\xi}dx = \int f(y)e^{-2\pi i y\xi}\Big(\int g(x-y)e^{-2\pi i(x-y)\xi}dx\Big)dy = \hat{f}(\xi)\hat{g}(\xi)$. ∎

**Theorem 1.12 (Plancherel)**

If $f \in \mathcal{S}(\mathbb{R})$, then $\|\hat{f}\| = \|f\|$, where $\|f\| = \big(\int |f|^2\big)^{1/2}$.

**Proof.** Let $f^\flat(x) = \overline{f(-x)}$; then $\widehat{f^\flat}(\xi) = \overline{\hat{f}(\xi)}$. With $h = f * f^\flat$, $\hat{h} = \hat{f}\overline{\hat{f}} = |\hat{f}|^2$ (Proposition 1.11) and $h(0) = \int |f|^2$. By Fourier inversion at 0, $h(0) = \int \hat{h}(\xi)d\xi = \int |\hat{f}|^2$. ∎

### 5.1.7 Extension to functions of moderate decrease

For $f, g$ of moderate decrease, $f * g$ is of moderate decrease and $\widehat{f*g} = \hat{f}\hat{g}$; the multiplication formula still holds. Hence Fourier inversion and Plancherel hold whenever $f$ and $\hat{f}$ are both of moderate decrease.

### 5.1.8 The Weierstrass approximation theorem

**Theorem 1.13 (Weierstrass)**

If $f$ is continuous on $[a,b]$ and $\epsilon > 0$, there is a polynomial $P$ with $\sup_{[a,b]} |f - P| < \epsilon$.

**Proof.** Choose $[-M, M] \supset [a,b]$ in its interior, and extend $f$ to a continuous $g$ on $\mathbb{R}$, linear from $f(b)$ to 0 on $[b, M]$, from $f(a)$ to 0 on $[-M, a]$, and 0 outside $[-M, M]$; $|g| \leq B$. $g$ is uniformly continuous, so as in Corollary 1.7, $g * K_\delta \to g$ uniformly; fix $\delta$ with $|g * K_\delta - g| < \epsilon/2$. Expand $e^{-\pi x^2/\delta} = \sum_n \frac{(-\pi x^2/\delta)^n}{n!}$, uniformly convergent on $|x| \leq 2M$, and truncate: a polynomial $R$ with $|K_\delta(x) - R(x)| < \frac{\epsilon}{4MB}$ for $|x| \leq 2M$. Then $P(x) = \int_{-M}^{M} g(t)R(x-t)\,dt$ is a polynomial in $x$, and for $|x| \leq M$ (so $|x - t| \leq 2M$),

$$|(g * K_\delta)(x) - P(x)| = \Big|\int_{-M}^{M} g(t)\big(K_\delta(x-t) - R(x-t)\big)dt\Big| \leq 2MB\frac{\epsilon}{4MB} = \frac{\epsilon}{2}.$$

∎

## 5.2 Applications to some partial differential equations

**Motivation.** The Fourier transform turns $\partial_x^2$ into multiplication by $-4\pi^2\xi^2$, so the heat equation on $\mathbb{R}$ is solved by convolving with a Gaussian, and Laplace's equation in the half-plane by convolving with the Poisson kernel. Uniqueness needs some control at infinity.

### 5.2.1 The time-dependent heat equation on the real line

Seek $u(x,t)$ with $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ $(t>0)$, $u(x,0)=f(x)$. Transforming in $x$: $\partial_t \hat{u}(\xi,t) = -4\pi^2\xi^2\hat{u}(\xi,t)$, so $\hat{u}(\xi,t) = \hat{f}(\xi)e^{-4\pi^2\xi^2 t}$, and $u = f * \mathcal{H}_t$ with the **heat kernel**

$$\mathcal{H}_t(x) = K_\delta(x) \text{ with } \delta = 4\pi t, \qquad \mathcal{H}_t(x) = \frac{1}{(4\pi t)^{1/2}} e^{-x^2/4t}, \qquad \widehat{\mathcal{H}_t}(\xi) = e^{-4\pi^2 t\xi^2}.$$

**Theorem 2.1**

Given $f \in \mathcal{S}(\mathbb{R})$, let $u(x,t) = (f * \mathcal{H}_t)(x)$ for $t>0$. Then:
(i) $u$ is $C^2$ for $x \in \mathbb{R}$, $t>0$, and solves the heat equation.
(ii) $u(x,t) \to f(x)$ uniformly in $x$ as $t \to 0$; hence, setting $u(x,0)=f(x)$, $u$ is continuous on $\{t \geq 0\}$.
(iii) $\int |u(x,t) - f(x)|^2 dx \to 0$ as $t \to 0$.

**Proof.**

(i) $u(x,t) = \int \hat{f}(\xi) e^{-4\pi^2\xi^2 t} e^{2\pi i x\xi} d\xi$ (inversion, Proposition 1.11). Differentiating under the integral (justified by the rapid decay of $\hat{f}$) gives $\partial_t u = \int \hat{f}(-4\pi^2\xi^2) e^{\cdots} = \partial_x^2 u$.

(ii) Corollary 1.7 with $\delta = 4\pi t$.

(iii) By Plancherel, $\int |u-f|^2 = \int |\hat{f}(\xi)|^2 |e^{-4\pi^2\xi^2 t} - 1|^2 d\xi$. Split at $|\xi| \leq N$ (where $|e^{-4\pi^2\xi^2 t} - 1| \leq 4\pi^2 N^2 t$) and $|\xi| > N$ (where it is $\leq 1$, and $\int_{|\xi|>N} |\hat{f}|^2 < \epsilon$). ∎

**Corollary 2.2**

$u(\cdot,t) \in \mathcal{S}(\mathbb{R})$ uniformly in $t$: for every $T>0$ and $k, \ell \geq 0$,

$$\sup_{x\in\mathbb{R},\ 0<t<T} |x|^k |\partial_x^\ell u(x,t)| < \infty. \tag{5.1}$$

**Proof.** $\partial_x^\ell u = (\partial_x^\ell f) * \mathcal{H}_t$, so it suffices to treat $\ell = 0$; let $|x| \geq 1$. Split $\int f(x-y)\mathcal{H}_t(y)dy$ into $|y| \leq |x|/2$ and $|y| \geq |x|/2$. On the first part $|x-y| \geq |x|/2$, so $|f(x-y)| \leq C_k |x|^{-k}$, and $\int \mathcal{H}_t = 1$. On the second part $\mathcal{H}_t(y) \leq (4\pi t)^{-1/2} e^{-x^2/16t}$, and [Supplement] $\sup_{0<t<T} t^{-1/2} e^{-x^2/16t} \leq C_{k,T}|x|^{-k}$ (put $s = x^2/t \geq x^2/T$: $t^{-1/2}e^{-x^2/16t} = |x|^{-1} s^{1/2} e^{-s/16} \leq C|x|^{-1} e^{-s/32} \leq C|x|^{-1} e^{-x^2/32T}$); together with $\int |f| < \infty$ this gives $|u(x,t)| \leq C|x|^{-k}$. ∎

**Theorem 2.3 (Uniqueness)**

Suppose (i) $u$ is continuous on $\{t \geq 0\}$;
(ii) $u$ satisfies the heat equation for $t>0$;
(iii) $u(x,0) = 0$;
(iv) $u(\cdot,t) \in \mathcal{S}(\mathbb{R})$ uniformly in $t$ as in (5.1). Then $u = 0$.

**Proof.** Let $E(t) = \int |u(x,t)|^2 dx \geq 0$, $E(0) = 0$. Differentiating under the integral and integrating by parts (the decay kills boundary terms),

$$\frac{dE}{dt} = \int (\partial_t u\, \bar{u} + u\, \overline{\partial_t u}) = \int (\partial_x^2 u\, \bar{u} + u\, \overline{\partial_x^2 u}) = -2 \int |\partial_x u|^2 \leq 0.$$

So $E$ is decreasing, $E \equiv 0$ (by continuity at $t = 0$), and $u = 0$. ∎

($\mathcal{H}_t(x)$ itself, or $u$ with very fast growth, shows the role of (iv): without decay conditions uniqueness fails.)

### 5.2.2 The steady-state heat equation in the upper half-plane

Find $u$ on $\mathbb{R}^2_+ = \{(x,y) : y > 0\}$ with $\triangle u = 0$ and $u(x,0) = f(x)$. Transforming in $x$: $\partial_y^2 \hat{u} = 4\pi^2 \xi^2 \hat{u}$, so $\hat{u} = A(\xi) e^{2\pi|\xi|y} + B(\xi) e^{-2\pi|\xi|y}$; boundedness gives $\hat{u}(\xi, y) = \hat{f}(\xi) e^{-2\pi|\xi|y}$, and $u = f * \mathcal{P}_y$ with the **Poisson kernel** $\mathcal{P}_y(x) = \frac{1}{\pi} \frac{y}{x^2+y^2}$.

**Lemma 2.4**

$\int_{-\infty}^{\infty} e^{-2\pi|\xi|y} e^{2\pi i \xi x} d\xi = \mathcal{P}_y(x)$ and $\int_{-\infty}^{\infty} \mathcal{P}_y(x) e^{-2\pi i x \xi} dx = e^{-2\pi|\xi|y}$.

**Proof.** Split the first integral at 0: $\int_0^\infty e^{-2\pi\xi(y-ix)} d\xi + \int_0^\infty e^{-2\pi\xi(y+ix)} d\xi = \frac{1}{2\pi(y-ix)} + \frac{1}{2\pi(y+ix)} = \frac{1}{\pi} \frac{y}{x^2+y^2}$. Both $\mathcal{P}_y$ and $e^{-2\pi|\xi|y}$ are of moderate decrease, so inversion (§5.1.7) gives the second identity. ∎

**Lemma 2.5**

$\{\mathcal{P}_y\}$ is a family of good kernels on $\mathbb{R}$ as $y \to 0$.

**Proof.** $\mathcal{P}_y > 0$; $\int \mathcal{P}_y = \widehat{\mathcal{P}_y}(0) = 1$ (or $\frac{1}{\pi} \int \frac{dx}{1+x^2} = 1$ after $x \mapsto yx$); $\int_{|x|\geq\delta} \mathcal{P}_y = \frac{1}{\pi} \int_{|x|\geq\delta/y} \frac{dx}{1+x^2} \to 0$. ∎

**Theorem 2.6**

Given $f \in \mathcal{S}(\mathbb{R})$, let $u(x,y) = (f * \mathcal{P}_y)(x)$. Then:
(i) $u$ is $C^2$ in $\mathbb{R}^2_+$ and $\triangle u = 0$;
(ii) $u(x,y) \to f(x)$ uniformly as $y \to 0$;
(iii) $\int |u(x,y) - f(x)|^2 dx \to 0$ as $y \to 0$;
(iv) with $u(x,0) = f(x)$, $u$ is continuous on $\overline{\mathbb{R}^2_+}$ and vanishes at infinity: $u(x,y) \to 0$ as $|x| + y \to \infty$.

**Proof.**

(i)–(iii) As for Theorem 2.1: $u = \int \hat{f}(\xi) e^{-2\pi|\xi|y} e^{2\pi i x \xi} d\xi$ and $(\partial_x^2 + \partial_y^2) e^{-2\pi|\xi|y} e^{2\pi i x \xi} = 0$; Lemma 2.5; Plancherel.

(iv) Splitting the integral at $|t| \leq |x|/2$ and $|t| \geq |x|/2$: $|u(x,y)| \leq C(\frac{1}{1+x^2} + \frac{y}{x^2+y^2})$. Also $|u| \leq \sup \mathcal{P}_y \int |f| \leq C/y$. Use the first bound for $|x| \geq y$ and the second for $|x| \leq y$. ∎

**Theorem 2.7**

If $u$ is continuous on $\overline{\mathbb{R}^2_+}$, $\triangle u = 0$ in $\mathbb{R}^2_+$, $u(x,0) = 0$, and $u$ vanishes at infinity, then $u = 0$.

**Lemma 2.8 (Mean-value property)**

Let $u$ be $C^2$ with $\triangle u = 0$ in an open $\Omega \subseteq \mathbb{R}^2$. If the closed disc of radius $R$ centred at $(x, y)$ lies in $\Omega$, then $u(x, y) = \frac{1}{2\pi}\int_0^{2\pi} u(x + r\cos\theta, y + r\sin\theta)\, d\theta$ for $0 \le r \le R$.

**Proof.** Let $U(r, \theta) = u(x + r\cos\theta, y + r\sin\theta)$; in polar coordinates $0 = \partial_\theta^2 U + r\partial_r(r\partial_r U)$. With $F(r) = \frac{1}{2\pi}\int_0^{2\pi} U(r, \theta)d\theta$, $r\partial_r(r\partial_r F) = -\frac{1}{2\pi}\int \partial_\theta^2 U\, d\theta = 0$ (periodicity). So $rF'(r)$ is constant, equal to its value 0 at $r = 0$; $F$ is constant, $F(r) = F(0) = u(x, y)$. ■

**Proof of Theorem 2.7.** (The example $u = y$ shows the decay is needed.) Take $u$ real (real and imaginary parts) and suppose $u(x_0, y_0) > 0$. Choose $R$ so large that $u \le \frac{1}{2}u(x_0, y_0)$ outside the half-disc $D_R^+ = \{x^2 + y^2 \le R^2,\ y \ge 0\}$. $u$ attains its maximum $M$ on $D_R^+$ at some $(x_1, y_1)$, and $u \le M$ on all of $\overline{\mathbb{R}^2_+}$; $y_1 > 0$ since $u = 0$ on $y = 0$ and $M > 0$. By the mean-value property on discs of radius $r < y_1$, $M = u(x_1, y_1) = \frac{1}{2\pi}\int u(x_1 + r\cos\theta, y_1 + r\sin\theta)d\theta \le M$, so $u = M$ on all such circles (continuity), hence on the disc of radius $y_1$ around $(x_1, y_1)$. Its boundary touches the real axis, where $u = 0$: contradiction with $M > 0$. Similarly $u \ge 0$. ■

## 5.3 The Poisson summation formula

**Motivation.** Periodizing a function on $\mathbb{R}$ gives a function on the circle whose Fourier coefficients are samples of $\hat{f}$ at the integers. This links Fourier analysis on $\mathbb{R}$ and on the circle: it gives the functional equation of the theta function and identifies the heat and Poisson kernels of the circle as periodizations.

For $f \in \mathcal{S}(\mathbb{R})$ the **periodization** $F_1(x) = \sum_{n\in\mathbb{Z}} f(x + n)$ is 1-periodic and continuous (uniform convergence on compacts); another periodic function attached to $f$ is $F_2(x) = \sum_n \hat{f}(n)e^{2\pi inx}$.

**Theorem 3.1 (Poisson summation formula)**

If $f \in \mathcal{S}(\mathbb{R})$, then $\sum_{n\in\mathbb{Z}} f(x + n) = \sum_{n\in\mathbb{Z}} \hat{f}(n)e^{2\pi inx}$. In particular $\sum_n f(n) = \sum_n \hat{f}(n)$.

**Proof.** Both sides are continuous and 1-periodic, and the right side converges absolutely. The $m$-th Fourier coefficient of the left side is

$$\int_0^1 \sum_n f(x + n)e^{-2\pi imx}dx = \sum_n \int_n^{n+1} f(y)e^{-2\pi imy}dy = \int_{-\infty}^{\infty} f(y)e^{-2\pi imy}dy = \hat{f}(m),$$

interchanging sum and integral by uniform convergence. That of the right side is also $\hat{f}(m)$. By uniqueness of Fourier coefficients for continuous functions (Corollary 2.2 of Chapter 2, after rescaling to period 1), the two sides agree. ■

(The formula also holds when $f$ and $\hat{f}$ are of moderate decrease, with the same proof.)

### 5.3.1 Theta and zeta functions

The **theta function** is $\vartheta(s) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 s}$ for $s > 0$.

**Theorem 3.2**

$s^{-1/2}\vartheta(1/s) = \vartheta(s)$ for $s > 0$.

**Proof.** Apply Poisson summation (at $x = 0$) to $f(x) = e^{-\pi s x^2}$, whose transform is $\hat{f}(\xi) = s^{-1/2} e^{-\pi \xi^2/s}$ (Corollary 1.5 with $\delta = 1/s$). ∎

This functional equation of $\vartheta$ yields, via $\pi^{-s/2}\Gamma(s/2)\zeta(s) = \int_0^\infty t^{s/2-1} \frac{\vartheta(t)-1}{2} dt$, the functional equation of the Riemann zeta function $\zeta(s) = \sum n^{-s}$ (Book II).

### 5.3.2 Heat kernels

**Theorem 3.3**

The heat kernel of the circle is the periodization of the heat kernel of the line: $H_t(x) = \sum_{n\in\mathbb{Z}} \mathcal{H}_t(x+n)$.

**Proof.** Poisson summation for $f = \mathcal{H}_t$, $\hat{f}(\xi) = e^{-4\pi^2 t\xi^2}$: $\sum_n \mathcal{H}_t(x+n) = \sum_n e^{-4\pi^2 t n^2} e^{2\pi i n x} = H_t(x)$. ∎

**Corollary 3.4**

$H_t$ is a good kernel as $t \to 0$ (on the circle $\mathbb{R}/\mathbb{Z}$, with integrals over $|x| \le \frac{1}{2}$).

**Proof.** $H_t > 0$ by Theorem 3.3, and $\int_{|x|\le 1/2} H_t = \hat{H}_t(0) = 1$. For $|x| \le \frac{1}{2}$, $H_t(x) = \mathcal{H}_t(x) + E_t(x)$ with $E_t(x) = \sum_{n\neq 0} \mathcal{H}_t(x+n) \le C t^{-1/2} \sum_{n\ge 1} e^{-cn^2/t} \to 0$ uniformly as $t \to 0$ ($|x+n| \ge |n|/2$). For $\delta \le |x| \le \frac{1}{2}$, $\mathcal{H}_t(x) \le (4\pi t)^{-1/2} e^{-\delta^2/4t} \to 0$. So $\int_{\delta\le|x|\le 1/2} H_t \to 0$. ∎

### 5.3.3 Poisson kernels

**Theorem 3.5**

$P_r(2\pi x) = \sum_{n\in\mathbb{Z}} \mathcal{P}_y(x+n)$ with $r = e^{-2\pi y}$.

**Proof.** Poisson summation for $f = \mathcal{P}_y$, $\hat{f}(\xi) = e^{-2\pi|\xi|y}$ (both of moderate decrease): $\sum_n \mathcal{P}_y(x+n) = \sum_n e^{-2\pi|n|y} e^{2\pi i n x} = \sum r^{|n|} e^{in(2\pi x)} = P_r(2\pi x)$. ∎

## 5.4 The Heisenberg uncertainty principle

**Motivation.** A function and its Fourier transform cannot both be sharply localized: the product of their spreads is at least $1/16\pi^2$, with equality only for Gaussians. In quantum mechanics this is the uncertainty relation between position and momentum.

**Theorem 4.1 (Heisenberg)**

If $\psi \in \mathcal{S}(\mathbb{R})$ with $\int |\psi|^2 = 1$, then

$$\Big(\int x^2 |\psi(x)|^2 dx\Big)\Big(\int \xi^2 |\hat{\psi}(\xi)|^2 d\xi\Big) \ge \frac{1}{16\pi^2},$$

with equality iff $\psi(x) = Ae^{-Bx^2}$ with $B > 0$ and $|A|^2 = \sqrt{2B/\pi}$. More generally $\big(\int (x-x_0)^2|\psi|^2\big)\big(\int (\xi-\xi_0)^2|\hat{\psi}|^2\big) \ge \frac{1}{16\pi^2}$ for all $x_0, \xi_0 \in \mathbb{R}$.

**Proof.**

*Step 1.* Integrating by parts ($\psi, \psi'$ rapidly decreasing),

$$1 = \int |\psi|^2 = -\int x \frac{d}{dx}|\psi|^2 = -\int (x\psi'\bar{\psi} + x\psi\overline{\psi'}),$$

so by Cauchy–Schwarz $1 \le 2\int |x||\psi||\psi'| \le 2\big(\int x^2|\psi|^2\big)^{1/2}\big(\int |\psi'|^2\big)^{1/2}$.

*Step 2.* By Proposition 1.2(iv) and Plancherel, $\int |\psi'|^2 = 4\pi^2 \int \xi^2 |\hat{\psi}|^2$. Squaring gives the inequality.

*Step 3 (equality).* Equality in Cauchy–Schwarz forces $\psi'(x) = \beta x \psi(x)$ (and the real-part step to be equality), so $\psi = Ae^{\beta x^2/2}$; $\psi \in \mathcal{S}$ forces $\beta = -2B < 0$, and $\int |\psi|^2 = |A|^2\sqrt{\pi/2B} = 1$.

*Step 4.* The general form follows by applying the inequality to $e^{-2\pi i x\xi_0}\psi(x + x_0)$ and changing variables. ∎

**Quantum mechanics.** For a state function $\psi$ with $\int |\psi|^2 = 1$, the probability that the particle lies in $(a, b)$ is $\int_a^b |\psi|^2$; the expected position is $\bar{x} = \int x|\psi|^2$ and the variance $(\Delta x)^2 = \int (x - \bar{x})^2 |\psi|^2$. The momentum is governed in the same way by $\hat{\psi}$ (frequency $\xi$ corresponds to momentum $h\xi$, $h$ Planck's constant), and Theorem 4.1 becomes $(\Delta x)(\Delta\xi) \ge \frac{1}{4\pi}$, i.e. $(\Delta\,\text{position})(\Delta\,\text{momentum}) \ge \frac{h}{4\pi}$.